

Evaluation of Lightweight and efficient Object Detectors for Small Underwater trash Detection in AUV Systems

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Abstract—Autonomous Underwater Vehicles (AUVs) that are responsible for environmental monitoring and trash removal will require effective visual recognition. However, accurately detecting small and opaque objects in dark marine environments continues to be a significant challenge, especially when considering the extremely limited power and computational constraints of marine edge hardware. Existing accurate models tend to be too computationally expensive for real-time deployment, while lightweight detectors struggle with tiny features. To address this gap, we evaluate three state-of-the-art object detection architectures YOLOv8, Faster R-CNN with a ResNet50-FPN backbone, and Faster R-CNN with a MobileNetv3-FPN backbone on Trash-ICRA19 dataset designed to reflect the low-visibility and low-contrast conditions encountered by AUVs. The ResNet50-FPN model performs well for localizing with mAP50-95 0.7166, but it fails to perform a well in classifying, with high recall but low precision on classes like plastic with F1 score 0.74 and ROV targets with F1 score 0.54. On the other hand, the MobileNetv3-FPN variant has much better classification stability with micro F1 0.918 while maintaining decent detection accuracy with mAP50-95 0.6569. This makes it better for platforms with limited resources. YOLOv8 achieves the best generalization results, obtaining high precision and recall 0.961 and 0.964, respectively and the highest mAP50-95 score 0.813, while maintaining highly efficient for real-time inference.

Index Terms—Underwater object detection, YOLOv8, ResNet50, MobileNetv3, Faster R-CNN

I. INTRODUCTION

The rapid increase in ocean pollution, specifically plastics and trash in ocean and underwater environments, results in more study interest into implementing AUVs to develop automatic detection systems. These systems are able to conduct large-scale, transparent, highly efficient recording data in complicated underwater environments, allowing essential help with protecting the environment and sustainable ocean exploration[1]. However, low visibility, color distortion, light reflec-

tions, and opaque areas that reduce image quality making it difficult to detect underwater objects[2]. Deep learning-based approaches appears to be appropriate for visual recognition as the previously mentioned nonlinear distortions are challenging for standard image-processing-based detection algorithms to handle.

Deep neural networks, specifically convolutional architectures, changed object detection and achieved outstanding results in outdoor vision tasks. Both above ground and underwater applications make extensive use of Faster R-CNN[3], the YOLO series[4], and EfficientDet[5]. However, models tailored for low light, high noise, and resource limited scenarios are required due to additional constraints caused by underwater conditions. Because AUVs usually run on embedded GPU systems like the NVIDIA Jetson Orin or TX2, where real-time inference performance is crucial, their installation on AUVs is frequently limited by power and computational constraints[6].

Recent developments in the lightweight convolutional backbone, including MobileNetv3 and ResNet50-FPN, have provided practical approaches for effective built-in deployment in embedded devices which maintain computational cost and accuracy [7]. By using multiscale feature maps, the Faster R-CNN framework when used with these feature pyramid networks improves the detection of small-scale and hidden objects. However, considering all of these improvements, Faster R-CNN variants still have high inference delays, making them unsuitable for real-time underwater applications [8].

The most recent architecture YOLOv8 combines separate head design, mosaic augmentation, and anchor-free detection. Both detection accuracy and generalization in complex environments are significantly improved in this approach. When contrasted with two-stage detectors YOLOv8 has shown exceptional robustness on several underwater datasets, such

as Trash-ICRA19 achieving the highest mAP and inference speeds. Indeed these enhancements render YOLOv8 particularly valuable, in AUV-based marine monitoring, where both precision and efficiency're essential.

The Trash-ICRA19 dataset addresses an important need in comparing results for underwater trash detection algorithms. It contains realistic underwater environments with dark lighting, cluttered backgrounds, and limited visibility that accurately depict true underwater rubbish situations. However, models trained on this dataset frequently demonstrate an imbalance in either spatial localization or classification precision, especially when it comes to the detection of small or transparent trash like nets and plastic bottles. These constraints subsequently result in the requirement of careful analysis of the current designs in order to establish the optimal solution among detection accuracy and processing efficiency for embedded underwater devices.

II. RELATED WORK

Underwater object detection has become as one of the most wanted tasks over the last few years, with increasing requirements for autonomous systems that will be able to recognize and categorize underwater objects. In general, Deep learning architectures—and CNNs in particular—have greatly impacted the field of underwater computer vision by outperforming conventional image processing and feature-engineering approaches. However, particular issues such color loss, light dispersion, and uneven lighting make feature extraction difficult and decrease object visibility in underwater environments [8].

Domain-specific datasets such as Trash-ICRA19, TrashCan 1.0, and JAMSTEC underwater debris sets have made research in marine vision convenient. These datasets are mock-ups of real underwater scenes with different visibility, depth, and lighting conditions and could be used to test object detectors for low-visibility conditions, as tested by Saji et al. (2024)[1]. Among these, the Trash-ICRA19 dataset has very realistic low-contrast images, thus making it a standard benchmark for trash detection in marine robotics.

Two-stage object detectors, represented by the Faster R-CNN and Mask R-CNN, are widely used in underwater detection tasks given their high precision and robustness in localization. By combining hierarchical features, the Faster R-CNN with a ResNet50-FPN backbone can efficiently capture multiscale spatial data and achieve good localization performance even in low light. However, these architectures are not appropriate for real-time deployment on AUVs due to their high latency and high computational cost. A recent work by Sinthia (2023)[9] utilized MobileNetv3-FPN in the Faster R-CNN framework, attaining higher prediction speed with negligible loss in precision; thus, this detector has proven promising for embedded hardware such as the NVIDIA Jetson Orin and TX2.

The existing work by Pratima Sarkar[10] has proposed a Advanced YOLOv4 an improved version of YOLOv4 trained

for underwater imagery. The main contributions include underwater image enhancement which restores color, contrast, and sharpness in poor underwater photos using a U-Shape Transformer. Small object focused data augmentation which provides U-Mosaic, a modified Mosaic augmentation that makes training images contain more little items this improves the model's capacity to capture small-object characteristics. Improved Feature Aggregation via Double SPP which enhances the YOLOv4 neck by adding additional Spatial Pyramid Pooling(SPP) layer. This improves the identification of tiny and closely spaced objects by offering deeper multiscale feature fusion. Better Large-Object Handling which adds an additional CBL block and a down sampling step to the 13x13 prediction head. This helps in the detecting process by allowing very huge items to fit inside anchor boxes.

Lijia Guo [11] has proposed DeformableConvModule-YOLO (DCM-YOLO). The main contributions include Joint Image Enhancement Module(UnitModule) an improvement block that fixes low contrast, color cast, and blur in underwater images. Instead of operating as a separate preprocessor, it collaborates with the detector (shared loss). This uses a technique for estimating atmospheric light plus adaptive transmission. DCNv4-based Backbone which replaces standard convolutions with Deformable Convolution v4, enabling adaptive receptive fields, particularly useful for extracting small-object features and irregular biological forms and enhances small target sensitivity by combining D-LKA (deformable large kernel attention) and SPD (space-to-depth). Separated and Enhancement Attention Module (SEAM) which designed to handle occlusion and dense object crowds, improving robustness to overlapping organisms. Small Object Detection Head (SODH) which adds an additional prediction head at higher resolution to increase recall for tiny marine organisms.

Dehua Zhang [12] has proposed PRCII-Net, a lightweight underwater object detection framework designed to improve small object recognition while keeping computational demands low. The main contributions include Enhancing small-object detection in which new feature-interaction and feature-fusion modules on Marine datasets such as DUO contain over 98% small and medium objects, with dense clusters of sea urchins and starfish. Lightweight design for real-time inference the network uses 1x1 convolutions to compress channels and maintains high accuracy despite lowering FLOPs and parameters. PR-AIFI Attention Block (PR-AIFI) which applies multi-head self-attention only at low spatial resolution to model global context cheaply. CII-FPN (Dual-branch Cross-scale Fusion) SIAF introduces shallow high-resolution features for small-object sensitivity. DIAF fuses deep multi-resolution features, improving robustness to clustering.

Linlin Liu [13] has proposed a MarineYOLO network, an novel deep learning method to identify small recognition in underwater. The main contributions include EC2f module which enhances the capacity to retrieve fine-grained characteristics, which is crucial for small marine targets, and broadens the receptive field. Efficient Multi-scale Attention (EMA) strengthens the ability of network to recognize tiny, low-

contrast objects by assisting it in fusing spatial information across various scales. Channel plus Spatial Attention(CBAM) to concentrate on the most relevant feature areas for underwater sceneries, it was added to the FPN and head layers. Wise-IoU Loss which replaces CIoU to enhance localization stability for small targets and lessen gradient competition from poor anchors.

The evolution of one-stage detectors from YOLO, further advanced by YOLOv5, YOLOv7, and YOLOv8, has revolutionized real-time underwater detection. In order to improve precision-recall balance with light computation, YOLOv8 uses an anchor-free detection head, C2f cross-stage partial connections, and improved mosaic augmentation. In the evaluation of YOLOv8, EfficientDet-D0, and Faster R-CNN, Walia et al. (2025)[14] shown that YOLOv8 achieves a mAP of 0.96 at sub-80 ms inference latency, which would be ideal for onboard AUV deployment. Moorton et al. (2025)[2] done a detailed study of maritime trash recognition systems and further confirmed the superiority of YOLOv8 across several datasets such as TrashCan 1.0 and Trash-ICRA19.

These models demonstrate a trade-off of low precision for computational efficiency, which becomes important for operations that utilize constrained hardware without having access to high-performance GPUs. Comprehensive reviews by Samanth et al. (2025) [8] and Moorton et al. (2025)[2] highlight the major gaps in performance in underwater object detection, which include poor recognition of small invisible trash, a struggle to differentiate among cluttered objects well, and lack of temporal consistency in video-based detection. In essence, both YOLOv8 and Faster R-CNN have continued to dominate the environment of photographic identification, but their ability to adapt for dynamic underwater video feeds and multi-object scenarios remains an open subject for research[15].

III. METHODOLGY

A. Dataset Overview

The Trash-ICRA19[16] dataset, a publicly accessible benchmark for evaluating classification models in difficult underwater conditions, is used throughout these studies. This dataset was created to encourage the improvement of computer vision algorithms that can detect both oceanic trash and biologically relevant objects under low visibility, light dispersion, and colour delay conditions, characteristics commonly seen in underwater imagery. It has been recognized by many as one of the most precise and realistic underwater garbage detection datasets, therefore providing as a perfect match for the testing of deep learning architectures such as YOLOv8 and Faster R-CNN versions.

The dataset Trash-ICRA19 contains 7,684 labeled images collected under various underwater conditions using ROVs and other imaging platforms. The images represent multiple levels of visibility, reflections, and hiding of objects; therefore, ensuring an efficient analysis of detection and classification models. Each image is labeled with bounding boxes and category labels corresponding to three object classes

- **Biological Objects:** These include organisms and biological materials like marine plants, corals, and fish. Such objects commonly introduce noise in the tasks of trash detection on account of their rich textures and natural motion.
- **ROV:** This corresponds to any mechanical/man-made underwater gadget that would serve as hard challenges to train robust detection models.
- **Plastic Debris:** The primary focus of marine pollution detection is synthetic items such as bottles, containers, bags, and other pieces.

The data was divided into three subsets to make it easier model training and validation, Training set with 5,720 images, Validation set with 820 images, Test set with 1,144 images. This splitting ensures that each subset has a representation of the three categories in a balanced approach and maintains environmental diversity: deep surroundings, color, and brightness. The training set was utilized for the optimization of parameters, the validation set for model adjustments as well as early stopping, while the test dataset was used to evaluate the efficiency through metrics such as mAP50–95, Precision, Recall, and F1-score. Similar to earlier underwater detection research that benchmarked YOLOv8, Faster R-CNN (ResNet50-FPN), and Faster R-CNN (MobileNetv3-FPN) for marine trash detection, this study makes use of the Trash-ICRA19 dataset. Because of the dataset's wide range of situations, it serves as a crucial foundation for evaluating the robustness of detection and the dependability of classification for AUV applications with limited resources.

B. Data Preprocessing

A group of underwater-specific data augmentation strategies have been implemented to make the object detection models more reliable and enhance their generalization capabilities. The augmentation approaches were intended to represent the visual noise, illumination variations, and optical losses that one would usually experience in realistic marine scenarios. Standard augmentation methods are generally not effective enough for underwater vision because of the nonlinear absorption, scattering, and spectral distortion induced by water's physical properties. Thus, both photometric and geometric changes were conducted to better the responsiveness of the models to realistic field circumstances.

- **Geometric and Spatial Augmentations:** Basic geometric transformations were applied to enhance spatial consistency and to compensate for changes in viewpoint of underwater objects. Horizontal Flipping was randomly applied, $p=0.5$, to make the model adaptive to mirrored orientation. Mosaic Augmentation was applied by combining four randomly chosen photos into a single large composite, multi-context visual signals were added to improve the recognition of small and overlapping objects. This augmentation type has been helpful for numerous contemporary types of YOLO applications [14].
- **Photometric Augmentations:** Several color-based changes were implemented to mimic underwater optical

effects. Underwater Color Shift was applied to reduce the imbalance of light absorption by enhancing blue and green channels while reducing the red channel. And then CLAHE Enhancement which adds contrast constrained adaptive histogram equalization to the L-channel of the LAB color space, improving local contrast while avoiding the over-amplification of noise. And Underwater fog simulation which adds a semi-transparent white overlay to imitate turbidity and backscatter, prevalent in underwater imaging when images are recorded near suspended sediments; fog strength varies from 0.3 to 0.7. HSV Shift which simulates variations in light sources, water clarity, and color temperature by randomly varying hue and saturation. Gaussian Blur which applies a tiny kernel blur to approximate the loss of spatial clarity owing to particle dispersion and camera blurred motion from AUV or ROV.

C. Faster R-CNN with ResNet50-FPN backbone

Faster R-CNN was chosen as one of the benchmark detection models because of its proven capability to attain high accuracy in object localization and classification tasks, particularly when training is done on limited but high-quality data. For multiscale feature extraction, the architecture combines a Feature Pyramid Network with a ResNet50 backbone. This makes it easier to recognize small, low-contrast underwater trash in the Trash-ICRA19 dataset. This architecture consists of three major components

1) *Backbone Network (ResNet50-FPN)*: A pre-trained ResNet50 acts as the feature extractor, thereby giving high-level semantic characteristics from the input photos. The model will combine multiscale feature maps by integrating a Feature Pyramid Network (FPN), which improves the detection accuracy for small items like microplastics and ROV fragments. FPN combines features across many convolutional layers by leveraging lateral connections defined as

$$P_l = \text{Conv}_{1 \times 1}(C_l) + \text{Upsample}(P_{l+1}) \quad (1)$$

where C_l is the feature map from the l th convolutional stage of ResNet and P_l is the pyramid feature at level l . This formulation enables for high-resolution characteristics with detailed semantic context for region proposal generation.

2) *Region Proposal Network (RPN)*: The RPN generates suitable object regions by sliding a tiny window over the FPN feature maps. Bounding box dimensions and an objectness score are predicted by each anchor window.

3) *ROI Head and Predictor*: ROI features are pooled using ROIAlign, followed by fully connected layers that produce the final class probabilities and refined bounding box predictions. The total detection loss is given by :

$$L_{\text{total}} = L_{\text{cls}} + L_{\text{bbox}} \quad (2)$$

where L_{cls} represents the softmax cross-entropy loss for object classification, and L_{bbox} is the Smooth L1 loss applied to bounding box regression.

D. Faster R-CNN with MobileNetv3-FPN backbone

To overcome the power and computational restrictions of underwater edge devices deployed onboard AUVs, a light version of Faster R-CNN has been developed utilizing the backbone of MobileNetv3-Large with an FPN. This architecture enables real-time or close to real-time inference while maintaining decent detection capabilities under underwater visual settings, providing a very desirable trade-off between accuracy and efficiency.

MobileNetv3-FPN Faster R-CNN uses the same two-stage detection pipeline as ResNet50-FPN but replaces the complex convolutional layers of ResNet with depthwise separable convolutions and squeeze-and-excitation modules from MobileNetv3. These improvements significantly reduce the amount of parameters and FLOPs, making the model much more suited for embedded GPU architectures such as the NVIDIA Jetson Orin and TX2.

The backbone provides hierarchical feature maps which are fed into the Feature Pyramid Network (FPN) to produce semantically powerful, multi-scale representations:

$$P_l = \text{Conv}_{1 \times 1}(C_l) + \text{Upsample}(P_{l+1}) \quad (3)$$

where C_l is the feature map from the l th convolutional stage of ResNet and P_l is the pyramid feature at level l . This formulation enables for high-resolution characteristics with detailed semantic context for region proposal generation.

Faster R-CNN merges classification and regression losses across both the Region Proposal Network (RPN) and the ROI head, expressed as:

$$L_{\text{total}} = L_{\text{RPN}} + L_{\text{ROI}} = \sum_{i=1}^N [L_{\text{cls}}(p_i, p_i^*) + \lambda L_{\text{reg}}(t_i, t_i^*)] \quad (4)$$

where L_{cls} is the cross-entropy loss for predicted vs. ground-truth class labels, and L_{reg} is the Smooth L1 loss for the bounding box coordinates (t_i, t_i^*) , with $\lambda = 1.0$ as the weighting factor.

E. YOLOv8 Architecture

The YOLOv8 is a single-stage object detector—a model created by Ultralytics in 2023—was used as the high-performance baseline in the present study. Compared with the previous series of YOLO versions, namely YOLOv5 and YOLOv7, YOLOv8 features a number of major improvements: anchor-free detection, decoupled detection head, and better C2f (Cross Stage Partial) modules, balancing well between computational efficiency and representational capacity. With such features, YOLOv8 is particularly ideal for underwater object detection, which requires real-time inference and stability against illumination variations.

The YOLOv8 architecture consists of three core components: (i) a CSP-based backbone for feature extraction, (ii) a Path Aggregation Network (PAN) for multi-scale feature fusion, and (iii) a decoupled detection head for classification and regression.

1) *Backbone*: The YOLOv8 backbone is derived from the CSPDarknet family but replaces CSP blocks with C2f modules. Each C2f module concatenates all intermediate feature maps in order to improve the flow of gradients and capture both local and global context. The process of feature extraction can be expressed as:

$$F_{\text{out}} = \text{Concat}(F_{\text{in}}, \text{Conv}_{1 \times 1}(\text{Conv}_{3 \times 3}(F_{\text{in}}))) \quad (5)$$

where F_{in} and F_{out} denote the input and output feature maps, respectively. This structure reduces redundancy while maintaining strong feature representation.

2) *Neck*: The Path Aggregation Network (PAN) fuses low- and high-level features via top-down and bottom-up connections, enhancing multi-scale detection. The fused feature map P_l at each level l is calculated as:

$$P_l = \text{Conv}_{1 \times 1}(C_l) + \text{Upsample}(P_{l+1}) \quad (6)$$

where C_l is the feature map from the l th convolutional stage of ResNet and P_l is the pyramid feature at level l .

3) *Head*: Unlike previous YOLO versions, YOLOv8 adopts an anchor-free decoupled detection head which separates the classification and regression tasks, encouraging better convergence and stability in training. For every grid point, the network predicts class probabilities (p_c) and bounding box offsets (t_x, t_y, t_w, t_h)

IV. EXPERIMENTAL SETUP

All experiments were performed in the cloud platform Kaggle, which provided GPU-accelerated runtime environments suitable for deep learning tasks. The three detection models, Faster R-CNN (ResNet50-FPN), Faster R-CNN (MobileNetv3-FPN), and YOLOv8-m, were trained and evaluated using mixed-precision computation enabled with CUDA 12.1 on two NVIDIA T4 GPUs (16 GB VRAM each).

V. RESULTS AND ANALYSIS

Standard metrics were computed, using the COCO-style evaluation techniques, to evaluate the detection performance of the three models. These consist of Precision (P), Recall (R), F1-score (F1), mean Average Precision (mAP) at IoU thresholds of 0.5 and 0.5–0.95. The mAP@50–95 metric provides a more comprehensive measure of localization accuracy over widely varied overlap thresholds, whereas the F1-score balances classification precision with recall.

A. Results for Faster R-CNN (ResNet50-FPN)

The Faster R-CNN, ResNet50-FPN showed great localization performance, but it showed a lot of inconsistency in the classification, notably for those classes which had poor inter-class separability, such as plastic and ROV. Therefore, the model gave an overall mAP@50–95 = 0.7166, with mAP@50 = 0.9688, suggesting strong bounding box alignment but low precision.

Plastic objects had the maximum recall of 0.995, but their precision was only 0.59, primarily due to frequent misclassifications in cluttered backgrounds, according to per-class F1-scores. The ROV class received an F1 of 0.54, mostly owing to false positives produced by metallic or reflecting trash.

Strong spatial accuracy was shown by the mean IoU of 0.8615 across all matched combinations. Class-wise IoUs were as follows: plastic = 0.8545, biological = 0.8686, and ROV = 0.8733, further validating the model's robustness in object localization despite reduced accuracy.

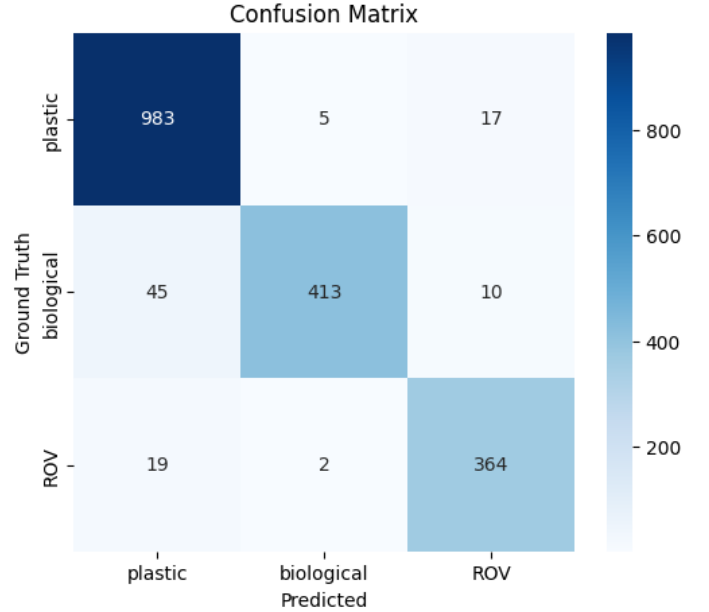


Fig. 1. Confusion Matrix of ResNet50.

B. Results for Faster R-CNN (MobileNetv3-FPN)

The Faster R-CNN with MobileNetv3-FPN variant shows significantly better classification stability while keeping decent detection accuracy. The model obtained mAP@50 = 0.9395 and mAP@50–95 = 0.6569, showing a little lower localization precision compared to ResNet50-FPN, but much higher efficiency and stability across categories.

At an IoU threshold of 0.5 and with a confidence of 0.4, the model obtained a micro-F1 score of 0.918, while the per-class F1-scores were 0.941 (plastic), 0.891 (biological), and 0.883 (ROV). The increase in performance illustrates that the MobileNetv3 backbone performs effectively when there are severe processing constraints—for instance, in low-power, real-time contexts—an important advantage for embedded AUV platforms.

Furthermore, the network's outstanding generalization in underwater environments was established by the confusion matrix's low false negative rate and total recall of 92.9% across all categories.

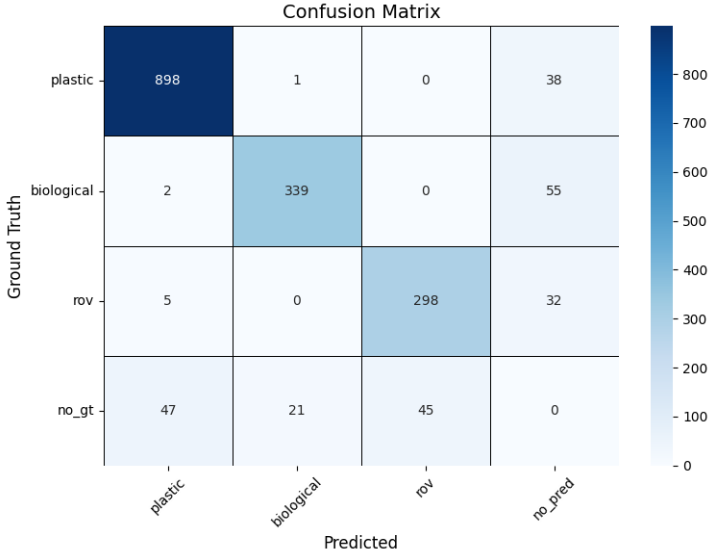


Fig. 2. Confusion Matrix of MobileNetV3 .

C. Results for YOLOv8-m

The YOLOv8-m model outperformed both Faster R-CNN variants in precision, recall, and overall mean AP, demonstrating exceptional performance across the board. An anchor-free detection paradigm, a decoupled classification-regression head, and an enhanced backbone architecture of C2f dramatically improved performance. Additionally, per-class performance was consistently high: ROV: 0.835, biological: 0.828, and plastic: $mAP@50-95 = 0.775$. These findings validate YOLOv8's capacity to provide high accuracy in small and semi-transparent debris detection under the demanding illumination circumstances typical of underwater images. The efficiency of feature learning and generalization under underwater conditions that were characterized by turbidity, illumination variance, and partial occlusions.

The model had a precision of 0.961, a recall of 0.964, $mAP@50 = 0.984$, and $mAP@50-95 = 0.813$, which established it as the top-performing architecture across all three frameworks under examination. Its per-class $mAP@50-95$ results further underscore the balanced detection performance of YOLOv8 for varied item categories: plastic, 0.775; biological, 0.828; and ROV, 0.835. This is suggestive of the robustness of YOLOv8 against class imbalance and intra-class variability that generally impair underwater detection systems.

This is because the model's anchor-free head eliminates reliance on pre-established anchor priors by directly regressing object centers and bounding box dimensions. This characteristic improves adaptability to irregular living species or deformable plastic debris, which are frequently encountered in natural underwater habitats. Furthermore, compared to conventional Smooth L1 or IoU losses, a CIoU-based bounding box loss function smoothly integrates spatial overlap, center distance, and aspect ratio alignment, resulting in tighter and more precise bounding box localization.

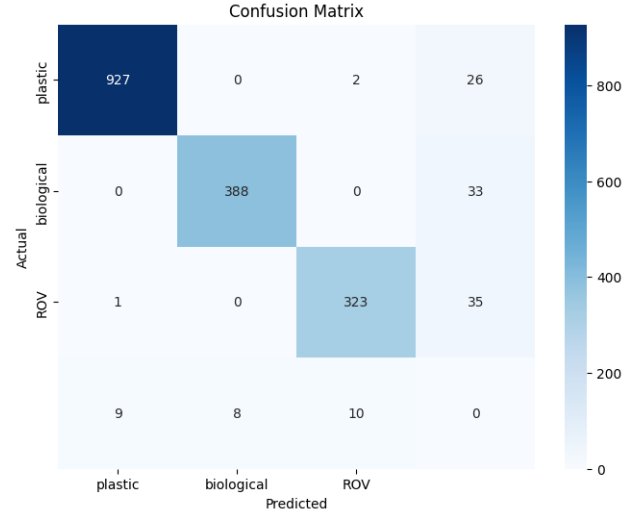


Fig. 3. Confusion Matrix of YOLOv8 .

D. Comparative Analysis

Quantitative results across the three architectures are described in Table 1. YOLOv8 maintains the strongest mAP across IoU thresholds while notably outperforming both Faster R-CNN variants in terms of overall detection accuracy and F1-score. While the ResNet50-FPN backbone delivered a significant boost in localization precision at the expense of decreased classification reliability, the MobileNetV3-FPN model offered the best trade-off between accuracy and computational efficiency.

Although the ResNet50-FPN model is exceptionally good at spatial localization, it is somewhat unstable for class-wise classification, largely due to substantial intra-class variation and feature redundancy. Because AUVs have limited processing capability, the MobileNetV3-FPN model provides classification stability with decreased complexity, making it appropriate for onboard inferences. The YOLOv8-m model clearly delivers the best performance in terms of precision, recall, and $mAP@50-95$, making it the most suited architecture for real-time underwater debris identification.

Overall, YOLOv8 displays higher adaptation and efficiency under underwater restrictions, which proves that YOLOv8 is acceptable for deployment in autonomous maritime monitoring systems.

TABLE I
COMPARATIVE RESULTS OF OBJECT DETECTION MODELS ON TRASH-ICRA19 DATASET

Model	Backbone	Precision	Recall	F1-score	mAP@50	mAP@50-95
Faster R-CNN	ResNet50-FPN	0.590	0.980	0.740	0.9688	0.7166
Faster R-CNN	MobileNetV3-FPN	0.907	0.929	0.918	0.9395	0.6569
YOLOv8-m	C2f Backbone	0.961	0.964	0.962	0.9842	0.8127

VI. TEST PREDICTIONS

Figures 4-6 demonstrate some sample detection outputs of the trained YOLOv8-m model on the test dataset of Trash-ICRA19. It can be observed that the model has properly

detected various underwater things, such as biological entities, plastic waste, and ROV components, with high confidence scores. This indeed represents its durability against low illumination and high turbidity in actual underwater scenarios.

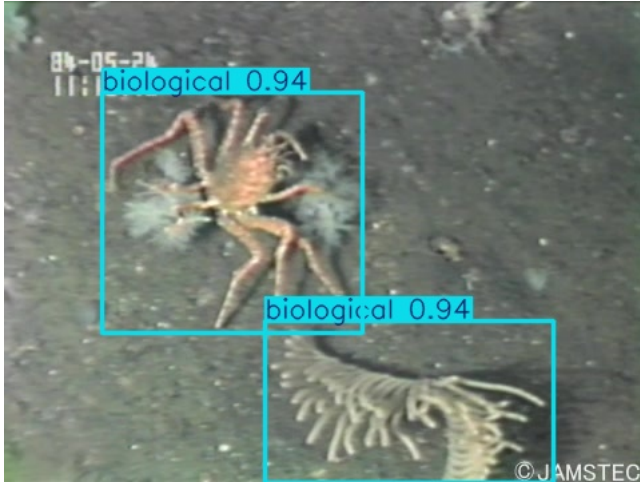


Fig. 4. Test Prediction-1.



Fig. 5. Test Prediction-2.

VII. CONCLUSION

This paper presents a comprehensive evaluation of state-of-the-art object identification architectures addressing underwater trash and environmental object recognition on a publically available dataset, specifically the garbage-ICRA19 dataset. Finding models that could provide high detection accuracy while still being computationally practical for AUVs and other embedded marine systems operating under stringent power and latency limitations was the aim.

Under identical experimental settings, three well-known architectures (Faster R-CNN with ResNet50-FPN, Faster R-CNN with MobileNetV3-FPN, and YOLOv8-m) were trained and evaluated using dual NVIDIA T4 GPUs. Each model was



Fig. 6. Test Prediction-3.

tested on its capacity to localize and categorize three target categories: plastic, biological things, and ROVs (Remotely Operated Vehicles), representing typical underwater visual problems.

These testing findings demonstrated that YOLOv8-m has outperformed both variations of Faster R-CNN for all quantitative criteria. It demonstrated a great capacity for accurate and dependable underwater object recognition with precision of 0.961, recall of 0.964, $mAP@50 = 0.984$, and $mAP@50-95 = 0.813$. Faster R-CNN with ResNet50-FPN demonstrated improved localization performance, $mAP@50 = 0.9688$, but decreased classification precision due to oversensitivity to background characteristics. On the other hand, the MobileNetV3-FPN backbone displayed higher class balance and computational efficiency, $F1 = 0.918$ and $mAP@50-95 = 0.6569$, demonstrating appropriate for light onboard processing.

Overall, results bring out a clear trade-off between detection accuracy, processing cost, and real-time inference capabilities. Faster R-CNN variations provide a more in-depth analysis and significant insights into the degrees of fine-grained feature extraction and high-quality localization, but their high inference latency limits its use on embedded hardware.

Future work will extend this framework toward multi-modal underwater sensing, integrating acoustic and LiDAR data with optical imagery to overcome visibility limitations. In order to further compress YOLOv8 architectures for their energy-efficient deployment on edge-AI processors in AUVs, methods such as knowledge distillation, quantization, and pruning will be tested in the future. Integration of self-supervised learning and domain adaptation may hold a potential for resolving the lack and variety of data across marine settings and, consequently, generalized underwater perception. In this work, it is shown that the state-of-the-art lightweight anchor-free detectors, such as YOLOv8, can achieve both high detection accuracy and real-time efficiency; thus, they could be an optimal choice for future autonomous marine vision systems to be used in preserving and monitoring underwater.

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